UCSD Spring 2026 DSC148: Data Mining

DSC 148: Intro to Data Mining – Data Mining Challenge

Dataset Description

Overview
This dataset contains Airbnb-style short-term rental listings from New York City. The goal is to predict the nightly listing price using structured listing attributes, host information, location fields, review statistics, booking rules, amenities, and free-text listing descriptions.
Each row represents one rental listing. The training file includes the target variable, price, while the test file contains the same listing information without price. Participants should build a regression model that predicts the nightly price for each listing in the test set.
Files
File Description
train.csv Training data with listing features and the target column price.
test.csv Test data with the same listing features as train.csv, excluding price.
mean_value_baseline.csv A simple baseline submission that predicts the mean training price for every test listing.
simple_linear_regression_baseline.csv A baseline submission generated from a simple linear regression model.
simple_baseline.ipynb Starter notebook showing how to load the data and create baseline predictions.

Prediction Task
This is a supervised regression problem.
Given information about a rental listing, predict the nightly rental price in dollars:
price
The training set contains 33,538 listings, and the test set contains 17,337 listings.
The average training price is approximately $145.18, with a median price of $109. Prices range from $0 to $1,999, so participants should be aware of possible outliers and skew in the target distribution.
Feature Summary
The dataset includes a mixture of numeric, categorical, date, boolean, and free-text fields. The major feature groups are summarized below.
Listing text fields
These fields describe the listing, neighborhood, rules, transit options, host interaction, and other narrative information.
Example columns: name, summary, space, description, neighborhood_overview, notes, transit, access, interaction, house_rules
Host information
These columns describe the host and their account. Some host-related fields contain missing values and may require cleaning or imputation.
Example columns: host_id, host_name, host_since, host_location, host_about, host_response_time, host_response_rate, host_is_superhost, host_neighbourhood, host_listings_count, host_verifications, host_has_profile_pic, host_identity_verified
Location fields
These fields describe where the listing is located. Listings are primarily located in New York City boroughs including Manhattan, Brooklyn, Queens, Bronx, and Staten Island.
Example columns: neighbourhood_cleansed, neighbourhood_group_cleansed, city, state, zipcode, market, country_code, country
Property and room attributes
These fields describe the physical rental unit and the type of stay being offered.
Example columns: property_type, room_type, accommodates, bathrooms, bedrooms, beds, bed_type, amenities, square_feet
Booking and policy fields
These columns describe booking rules, guest requirements, minimum and maximum stay lengths, and cancellation policy.
Example columns: guests_included, extra_people, minimum_nights, maximum_nights, instant_bookable, is_business_travel_ready, cancellation_policy, require_guest_profile_picture, require_guest_phone_verification
Review fields
These describe the number, timing, and quality of reviews. Review score fields contain missing values, especially for listings with few or no reviews.
Example columns: number_of_reviews, first_review, last_review, review_scores_rating, review_scores_accuracy, review_scores_cleanliness, review_scores_checkin, review_scores_communication, review_scores_location, review_scores_value, reviews_per_month
Column Common values
room_type Entire home/apt; Private room; Shared room
property_type Apartment; House; Townhouse; Loft; Condominium
bed_type Real Bed; Futon; Pull-out Sofa; Airbed; Couch

Submission Format
Submissions should contain two columns:
Id, Predicted
Column Description
Id The listing ID from test.csv.
Predicted The predicted nightly price.

Example submission rows:
Id Predicted
19307997 145.177291430616
20176193 145.177291430616

Notes for Participants
- Clean currency and percentage fields such as extra_people and host_response_rate before modeling.
- Encode categorical variables such as room_type, property_type, and neighbourhood_group_cleansed.
- Extract useful date features from host_since, first_review, and last_review.
- Handle missing values in review scores, host response fields, square_feet, and text columns.
- Engineer text features from listing descriptions, amenities, house rules, and neighborhood descriptions.
- Consider transforming the target with log(price + 1) to reduce skew, then converting predictions back to dollar values.
- Combine location, room type, property attributes, review history, host characteristics, and text-derived features for stronger performance.

Files
5 files

Size
158.04 MB

Type
csv, ipynb

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mean_value_baseline.csv (444.35 kB)

Detail Compact Column 2 of 2 columns

Id	Predicted
2539 – 29.8m	145 – 145
19307997	145.177291430616
20176193	145.177291430616
19485371	145.177291430616
13079990	145.177291430616
22339757	145.177291430616
7717071	145.177291430616
19416433	145.177291430616
28602814	145.177291430616
3167768	145.177291430616
22026277	145.177291430616
20982450	145.177291430616
16271468	145.177291430616
14445183	145.177291430616
6040356	145.177291430616
11484489	145.177291430616
29541600	145.177291430616
2793834	145.177291430616
588677	145.177291430616
22940375	145.177291430616
26126728	145.177291430616
8072662	145.177291430616
26250574	145.177291430616
19568889	145.177291430616
2048796	145.177291430616
24938213	145.177291430616
8555036	145.177291430616
22108302	145.177291430616
3625656	145.177291430616
11238715	145.177291430616
5041360	145.177291430616
3989564	145.177291430616
22662308	145.177291430616
17728208	145.177291430616
13597896	145.177291430616
13083638	145.177291430616
8760775	145.177291430616
25518930	145.177291430616
17256297	145.177291430616
1975069	145.177291430616
479285	145.177291430616
22050058	145.177291430616
21844210	145.177291430616
25835470	145.177291430616
27853108	145.177291430616
9486111	145.177291430616
23148095	145.177291430616
26603929	145.177291430616
6272181	145.177291430616
22674576	145.177291430616

Data Explorer
158.04 MB

- mean_value_baseline.csv
- simple_baseline.ipynb
- simple_linear_regression_base
- test.csv
- train.csv

Summary
- 5 files
- 133 columns

Download All